



Yassine Hallous

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Tunis, Sousse – Tunisia

PROFESSIONAL SUMMARY

A third-year Computer Engineering student specializing in Embedded Systems and IoT. Proven experience in app development, backend system design, and creating automated and smart solutions. Demonstrates strong leadership and project coordination skills through active involvement in IEEE.

ACADEMIC EXPERIENCE

- **PWN&PATCH**  Feb 2026 – May 2026
IoT Security Internship Menzah, Tunisia
 - Designed and implemented an **IoT Security Sandbox** to simulate devices, attacks, and defense strategies within a controlled environment.
 - Developed a scalable backend with **FastAPI**, **Celery**, and **Redis**, and deployed containerized IoT environments using **Docker** and **K3s**.
 - Integrated **Suricata IDS** and machine learning models (**XGBoost**, **Scikit-learn**) for real-time threat detection and anomaly classification
- **SofiaTech**  July 2025 – Aug 2025
Summer Internship Ariana, Tunisia
 - Developed **RuleGuard**, a Python tool for automated verification of C code against SOFIOS standards
 - Built a GUI for real-time analysis and reduced manual code review effort by 70%
- **SofiaTech**  July 2024 – Aug 2024
Summer Internship Ariana, Tunisia
 - Developed a Flutter mobile application integrated with ESP32 for real-time sensor monitoring
 - Implemented a PHP/MySQL backend for data storage and synchronization

EDUCATION

- **Higher Institute of Computer Science and Communication Technologies (ISITCOM)** September 2023 - June 2026
Bachelor's Degree in Computer Engineering Sousse, Tunisia
 - GPA: 16.63/20.00
- **Dar El Amen High School** 2023
Mathematics Baccalaureate Kairouan, Tunisia
 - GPA: 14.3/20.00

PROJECTS

- **SuperCleaner: A smart Chrome extension for web content cleaning.** December 2025
Tools: JavaScript, HTML, CSS, Manifest V3 
 - Built a Chrome extension to hide unwanted elements (ads, pop-ups, images, links, videos).
 - Added intelligent modes (Adult, Education, Image/Link/Icon/Video removal).
- **STL Web Viewer: 3D STL file viewer.** November 2025
Tools: React, TypeScript, Three.js, Tailwind CSS, Vite, Vercel 
 - Developed a web application to load and visualize large STL files.
 - Implemented smooth 3D rendering with Web Workers and optimized camera controls.
- **MNIST Digit Recognition: Handwritten Digit Classifier.** October 2025
Tools: Python, TensorFlow/Keras, Flask, React, TypeScript, Tailwind CSS 
 - Trained a Convolutional Neural Network (CNN) achieving 99.5% accuracy on MNIST dataset.
 - Built an interactive web application with React frontend and Flask API for real-time digit prediction.
- **CineMatch: An intelligent movie recommendation system.** October 2025
Tools: Python, Flask, Scikit-learn, React, TypeScript, Tailwind CSS 
 - Built a recommendation system using machine learning algorithms to suggest movies based on user interests.
 - Developed a modern web platform using React and TypeScript.

- **Nexova: An intelligent platform that automates industrial processes.** November 2025
Tools: Node.js, TypeScript, PostgreSQL, Electron 
 - Developed a backend system with TypeScript, Node.js, and PostgreSQL to process real-time IoT sensor data.
 - Designed RESTful APIs enabling seamless integration with frontend and IoT devices.
- **ExpenseTracker Mobile App:** April 2025
Tools: Java, XML, SQLite 
 - Built a mobile application for secure expense tracking using Java.
 - Integrated SQLite for offline data storage and efficient management.
- **QuizNight:** March 2025
Tools: Node.js, TypeScript, HTML, CSS, JavaScript 
 - Developed a full-stack quiz game platform with dynamic user interaction.
 - Implemented user management, scoring system, leaderboards, and custom quiz creation.
- **Journal App:** February 2025
Tools: Kotlin 
 - Created a mobile journal app using Kotlin, allowing users to add, view, and edit personal entries.
- **Messaging System:** March 2024
Tools: Linux, C 
 - Built a messaging server in C on Linux, enabling users to create accounts and exchange messages.

SKILLS

- **Programming Languages :** C, C++, Python, Java, Kotlin, Dart, JavaScript, TypeScript, PHP, HTML, CSS
- **AI & Machine Learning:** TensorFlow, Scikit-learn, Pandas, NumPy, MLflow, XGBoost
- **Mobile & Backend Development :** Flutter, Node.js, MySQL, PostgreSQL, MongoDB, InfluxDB, Firebase
- **DevOps & Cloud :** Docker, K3S, CI/CD, AWS
- **Embedded Systems & IoT :** ESP32, Arduino, Raspberry-Pi, MQTT
- **Tools & Platforms :** Git, GitHub, Postman, Figma, Matlab, Kali Linux, Ubuntu

HONORS AND AWARDS

- **2nd Place – Cyberbenders 3.0** January 2026
ATAST ISITCom & ATAST EPI 
 - Ranked 2nd with team **Bitwi\$e** in a CTF covering cryptography, web exploitation, OSINT, forensics, and AI-driven challenges
- **1st Place – IAS TAM Technical Challenge** October 2025
IAS Tunisia Annual Meeting 
 - Won first place with **Defensys**, an AI-powered platform for proactive application security
- **Best Android Project – GDG TechDays** April 2025
GDG Sousse 
 - Awarded best project in Android development bootcamp (Kotlin, mobile applications)
- **3rd Place – RedRoom Hackathon** March 2025
Securinets Hackathon 
 - Achieved 3rd place overall and 2nd in CTF during a 26-hour cybersecurity hackathon

VOLUNTEER EXPERIENCE

- **Chair of the IAS, IES, PES Joint Chapter** October 2023 - Present
IEEE ESSTHS Student Branch 
 - Organized and led technical bootcamps and industrial visits for student members.
 - Coordinated logistics and secured sponsorship for major events like Sight Day Congress and Robots League 2.0.
 - Contributed to event planning as a member of the TSYP 11 Organizing Committee.

CERTIFICATIONS

- **CompTIA: Security+ (In Progress)** Expected August 2026
- **Cisco Networking Academy: Python Essentials 2** July 2025
- **Cisco Networking Academy: Python Essentials 1** June 2025
- **TryHackMe: Pre-Security Learning Path** July 2025
- **Cisco Networking Academy: Introduction à la cybersécurité** June 2025
- **Cisco Networking Academy: Notions de base sur les réseaux** April 2025
- **IEEE: IEEEExtreme 18.0 Programming Competition** October 2024

ADDITIONAL INFORMATION

Languages: English (Fluent), French (Fluent), Arabic (Native)

Interests: Chess, Trading, Jiu jitsu